

49-amaliy mashg'ulot

Na'muna uchun misollar yechimi

1-misol. Ushbu

$$z = x^2 x^2 y + y^3$$

funksiyaning xususiy hosilalari topilsin.

Berilgan funksiyada y ni o'zgarmas deb qarab, x o'zgaruvchi bo'yicha hosilasini topamiz. Bunda ma'lum bo'lgan qoida va formulalardan foydalanamiz:

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(x^3 + x^2 y + y^3) = 3x^2 + 2xy + 0 = 3x^2 + 2xy$$

Huddi shunga o'xshash x ni o'zgarmas hisoblab topamiz:

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(x^3 + x^2 y + y^3) = 0 + x^2 + 3y^2 = x^2 + 3y^2$$

2-misol. Ushbu

$$f(x, y) = \sqrt{x^2 - y^2}$$

funksiyaning $(5; -3)$ nuqtadagi xususiy hosilalari topilsin.

Avvalo berilgan funksiyaning xususiy hosilalarini topamiz:

$$f'_x(x, y) = \frac{\partial}{\partial x}(\sqrt{x^2 - y^2}) = \frac{1}{2\sqrt{x^2 - y^2}} \cdot 2x = \frac{x}{\sqrt{x^2 - y^2}},$$

$$f'_y(x, y) = \frac{\partial}{\partial y}(\sqrt{x^2 - y^2}) = \frac{1}{2\sqrt{x^2 - y^2}} \cdot (-2y) = \frac{-y}{\sqrt{x^2 - y^2}}$$

Endi bu xususiy hosilalarning ko'rsatilgan $(5; -3)$ nuqtadagi qiymatlarini topamiz:

$$f'_x(5, -3) = \frac{5}{\sqrt{5^2 - (-3)^2}} = \frac{5}{\sqrt{25 - 9}} = \frac{5}{4},$$

$$f'_y(5, -3) = \frac{-(-3)}{\sqrt{5^2 - (-3)^2}} = \frac{3}{\sqrt{25 - 9}} = \frac{3}{4}$$

Demak, $f'_x(5; -3) = \frac{5}{4}$, $f'_y(5, -3) = \frac{3}{4}$ ►

3-misol. Ushbu

$$f(x) = f(x_1, x_2, \dots, x_m) = x_1^2 + x_2^2 + \dots + x_m^2$$

funksiyaning $\forall (x_1^0, x_2^0, \dots, x_m^0) \in R^m$ nuqtada differensiallanuvchi bo'lishi ko'rsatilsin.

◀ Berilgan funksiyaning $x^0 = (x_1^0, x_2^0, \dots, x_m^0)$ nuqtadagi to'liq orttirmasini topamiz:

$$\begin{aligned} \Delta f(x^0) &= (x_1^0 + \Delta x_1)^2 + (x_2^0 + \Delta x_2)^2 + \dots + (x_m^0 + \Delta x_m)^2 - \\ &- (x_1^{0^2} + x_2^{0^2} + \dots + x_m^{0^2}) = 2x_1^0 \Delta x_1 + 2x_2^0 \Delta x_2 + \dots + \\ &+ 2x_m^0 \Delta x_m + \Delta x_1^2 + \Delta x_2^2 + \dots + \Delta x_m^2. \end{aligned}$$

Agar

$$\begin{aligned} A_1 &= 2x_1^0, A_2 = 2x_2^0, \dots, A_m = 2x_m^0, \\ \alpha_1 &= \Delta x_1, \alpha_2 = \Delta x_2, \dots, \alpha_m = \Delta x_m \end{aligned}$$

deyilsa, unda

$$\Delta f(x^0) = A_1 \Delta x_1 + A_2 \Delta x_2 + \dots + A_m \Delta x_m + \alpha_1 \Delta x_1 + \alpha_2 \Delta x_2 + \dots + \alpha_m \Delta x_m$$

bo'ladi. Demak, berilgan funksiya $\forall x^0 \in R^m$ nuqtada differensiallanuvchi. ▶

4-misol. Ushbu

$$f(x, y) = \ln \operatorname{tg} \frac{x}{y}$$

funksiyaning xususiy hosilalar topilsin.

◀ Berilgan funksiyaning xususiy hosilalari quyidagicha bo'ladi:

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{\partial}{\partial x} \left(\ln \operatorname{tg} \frac{x}{y} \right) = \frac{1}{\operatorname{tg} \frac{x}{y}} \cdot \frac{1}{\cos^2 \frac{x}{y}} \cdot \frac{1}{y} = \frac{2}{y \sin \frac{2x}{y}}; \\ \frac{\partial f}{\partial y} &= \frac{\partial}{\partial y} \left(\ln \operatorname{tg} \frac{x}{y} \right) = \frac{1}{\operatorname{tg} \frac{x}{y}} \cdot \frac{1}{\cos^2 \frac{x}{y}} \cdot \left(-\frac{x}{y^2} \right) = \frac{-2}{y^2 \sin \frac{2x}{y}} \quad \blacktriangleright \end{aligned}$$

Misollar

Quyidagi funksiyalarning xususiylari hosilalari topilsin.

1. a) $z = x^3 + y^2 - 2xy$, b) $z = 5x^2 + 8xy^2 + y^3$

2. a) $z = x^3 + y^2 - 3axy$, b) $z = x^2 - 2xy + y^2$

3. a) $z = \frac{xy}{x+y}$, b) $z = \frac{x-y}{x+y}$

4. a) $z = \frac{y}{x}$, b) $z = \sqrt{\frac{x}{y}}$

5. a) $z = \sqrt{x^2 - y^2}$, b) $z = \sqrt{x+3y}$

6. a) $z = \frac{1}{\sqrt{x} - \sqrt{y}}$, b) $z = \frac{x}{\sqrt{x^2 + y^2}}$

7. Agar $f(x, y) = \frac{1-xy}{1+xy}$ bo'lsa, $f'_x(0,1)$, $f'_y(0,1)$ topilsin.

8. Agar $f(x, y) = \sqrt{xy + \frac{x}{y}}$ bo'lsa, $f'_x(2,1)$, $f'_y(2,1)$ topilsin.

9. Agar $f(x, y) = \ln \frac{x+y}{x-y}$ bo'lsa, $f'_x(2e, e)$, $f'_y(2e, e)$ topilsin.

10. Agar $f(x, y) = y \sin x + \cos(x-y)$ bo'lsa, $f'_x(\pi, 0)$, $f'_y(\pi, 0)$ topilsin.

Quyidagi murakkab funksiyalarning hosilalari topilsin.

1. Agar $z = e^{x^2+2y^2}$ bo'lib, $x = \sin t$, $y = \cos t$ bo'lsa, $\frac{dz}{dt}$ topilsin.

2. Agar $z = \frac{x}{y}$ bo'lib, $x = e^t$, $y = \ln t$ bo'lsa, $\frac{dz}{dt}$ topilsin.

3. Agar $z = \frac{x^2}{y}$ bo'lib, $x = u - 2v$, $y = v + 2v$ bo'lsa, $\frac{\partial z}{\partial u}$, $\frac{\partial z}{\partial v}$ lar topilsin.

4. Agar $z = \ln \sin \frac{x}{\sqrt{y}}$ bo'lib, $x = 3t^2$, $y = \sqrt{t^2 + 1}$ bo'lsa, $\frac{\partial z}{\partial t}$ topilsin.

5. Agar $z = x^2 - y^2$ bo'lib, $x = u \cos v$, $y = u \sin v$ bo'lsa, $\frac{\partial z}{\partial u}$, $\frac{\partial z}{\partial v}$ lar topilsin.

6. Agar $z = u^v$ bo'lib, $u = \sin x$, $v = \cos x$ bo'lsa, $\frac{\partial z}{\partial x}$ topilsin.

7. Agar $z = \ln \sqrt{\frac{u}{v}}$ bo'lib, $u = ax + by$, $v = ax - by$ bo'lsa, $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$ topilsin.
8. Agar $z = e^{\frac{y^2}{x}}$ bo'lib, $x = x(u, v)$, $y = y(u, v)$ bo'lsa, $\frac{\partial z}{\partial u}, \frac{\partial z}{\partial v}$ topilsin.